

Discussion Unit:

DSE Core, JMX, What
and Why

Introduce DSE JMX

- What does DSE Core JMX offer
- Why use DSE Core JMX



Discussion Lab:

Matching pairs – Match the attributes on the right with the areas on the left

DSE Core JMX, DSE
(Interfaces/Utilities)

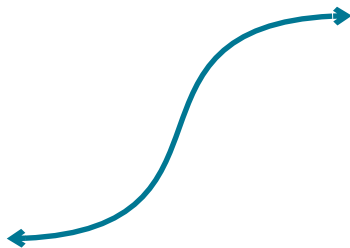
Discussion Lab: Which uses JMX



JMX



not-JMX



CQLSH



DSE Studio



Ops Ctr UI to Op Ctr Agent



Ops Ctr Agent to DSE Node



nodetool



DSBulk Loader



DSE Java Client Driver



dsetool



End of Discussion Lab:

DSE Core JMX, DSE
(Interfaces/Utilities)

Any Server Admin Communication Options-

<SOAP>



REST API



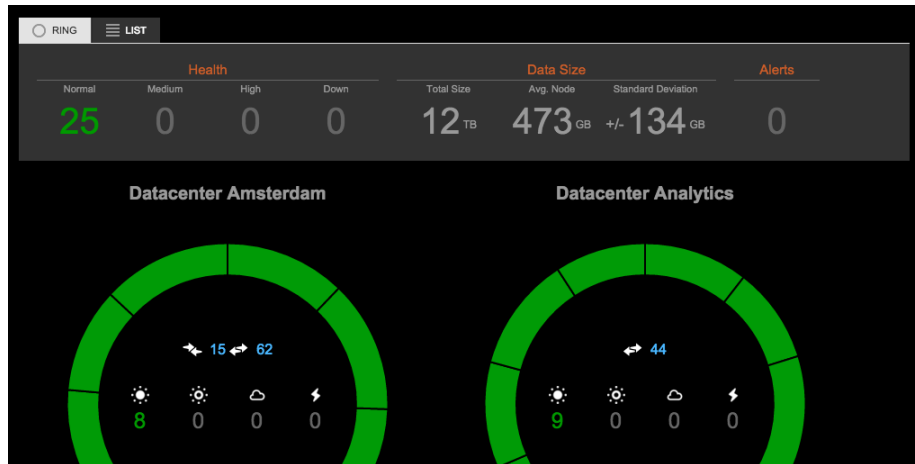
(others)

Is the server Java based ?

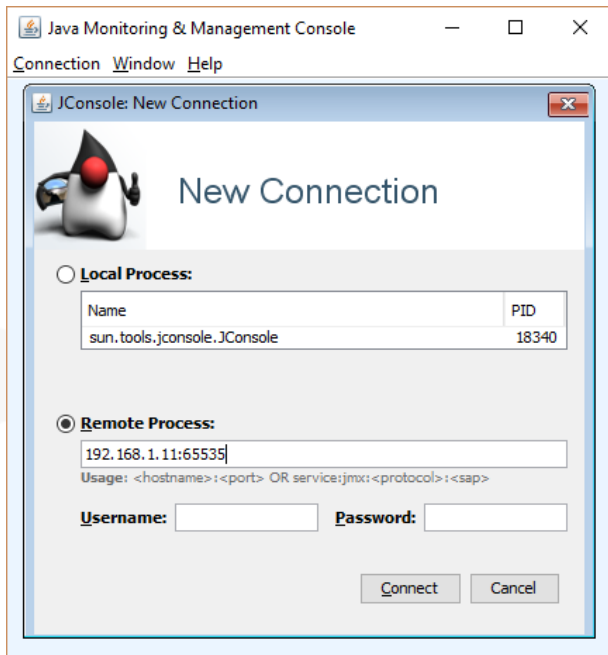
- Out of the box management of the JVM
- Pluggable/standard to most management agents, solutions
- References many existing Java interfaces; JNDI, others
- Standard NetBeans IDE interface
- Many existing clients

Getting (JMX like) data from DSE-

- DSE Ops Ctr; Web, GUI
- nodetool; reports on one node, command line, ASCII text
- dsetool
- CQL/System tables; Global (cluster wide view)
- JMX
 - Programmatic, customizable
 - Default install is local only, easily remote (security)



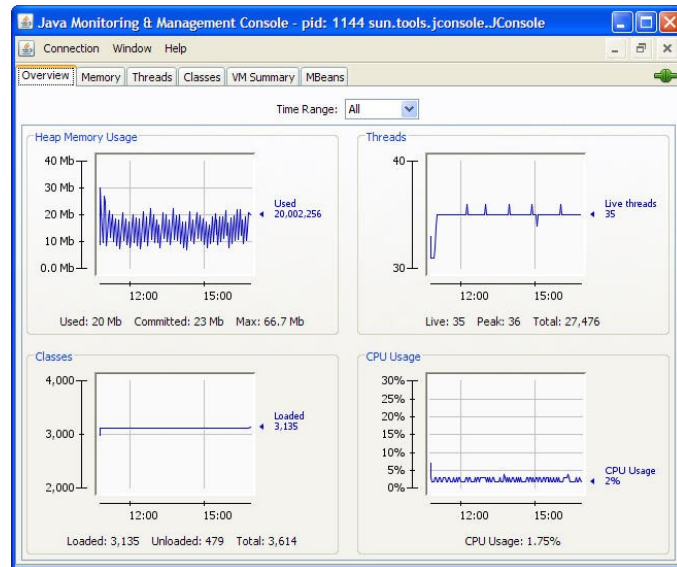
JMX, MBeans



- JMX: Java Management Extensions, standard way to manage a Java application
- Mbeans: Management Beans
 - Provided by the application
 - Attributes, can get() or set()
 - Operations, can invoke
 - Read *and* write
- Many client programs

JMX Clients: (many)

- nodetool, 80+ options, but formatted, documented, supported
- nodetool sjk
- JConsole
 - In Java SDK since 1.5
 - GUI, graphs too
- mx4j, JMX via Http
- Jolokia, JMX via Http/JSON
- jmxsh, command line (REPL)
- jmxterm, command line
- VisualVM



Securing (DSE) JMX-

JMX

jmx-port 7444

jmx-connection-type

- Local JMX - No Authentication
- ✓ Remote JMX - No Authentication
- Remote JMX - DSE Unified Authentication

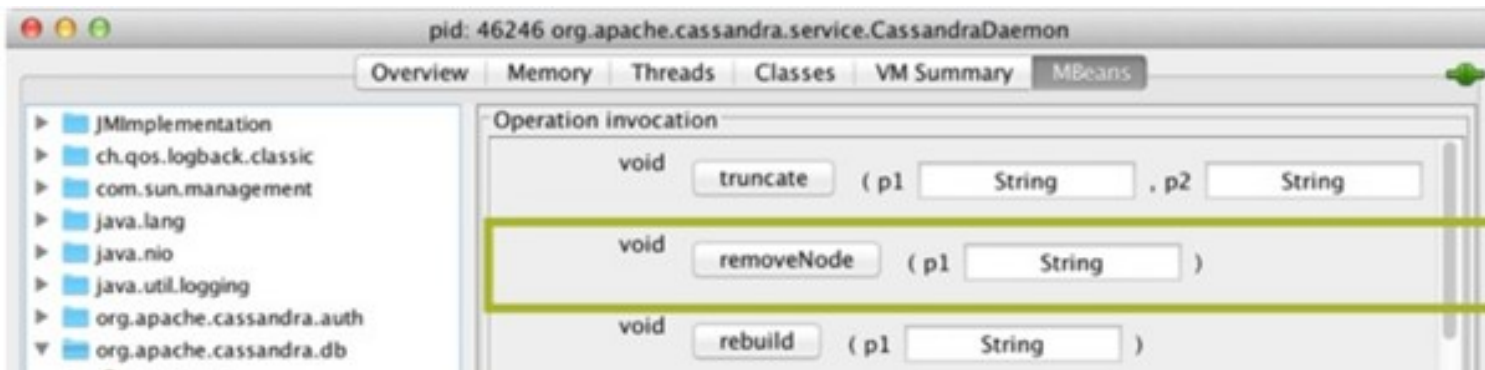
Configures the manner in which JMX will be exposed by DSE. Using "Remote JMX - No Authentication" is not recommended as this option allows unauthenticated remote connections to DSE.

jmx-remote-ssl ☐

- Via Ops Center
- Or Manually
- Super important
 - Consider mandatory for production
 - And when allowing remote clients

Why Secure JMX ?

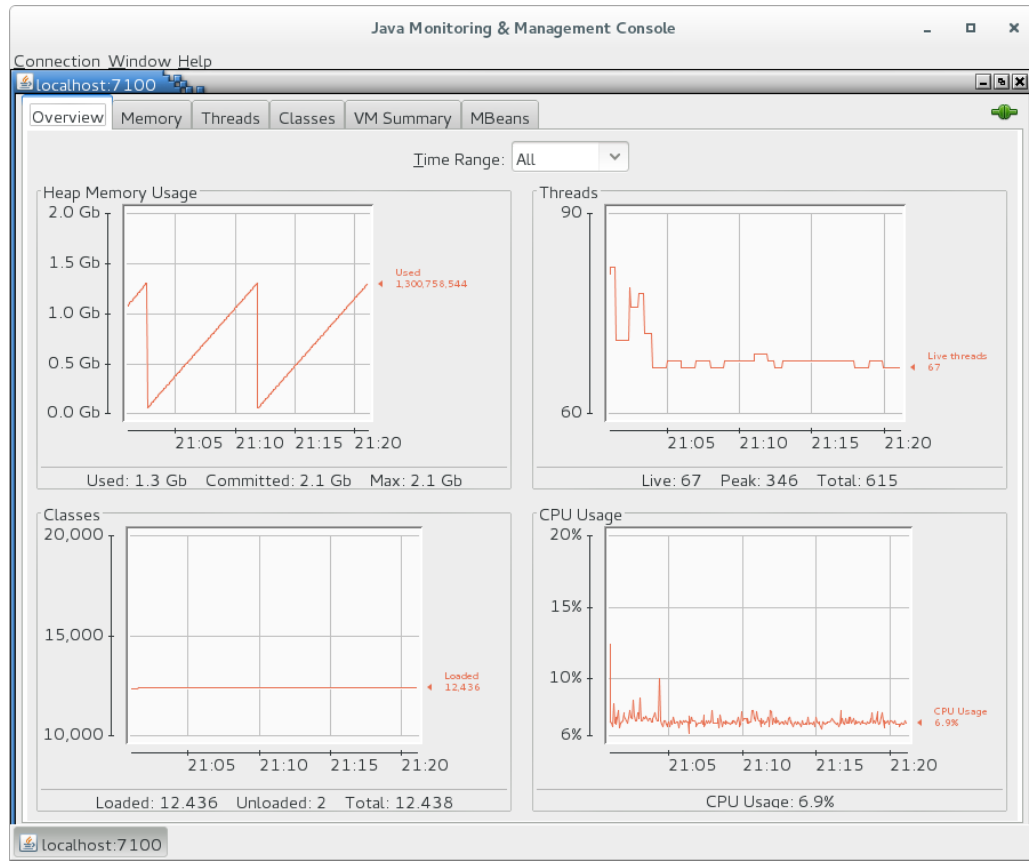
```
StorageService.removeNode( "coworker_IPADR" )
```



```
StorageService.truncate( "your_keyspace", "your_table" )
```

Using JConsole: jconsole localhost:7100

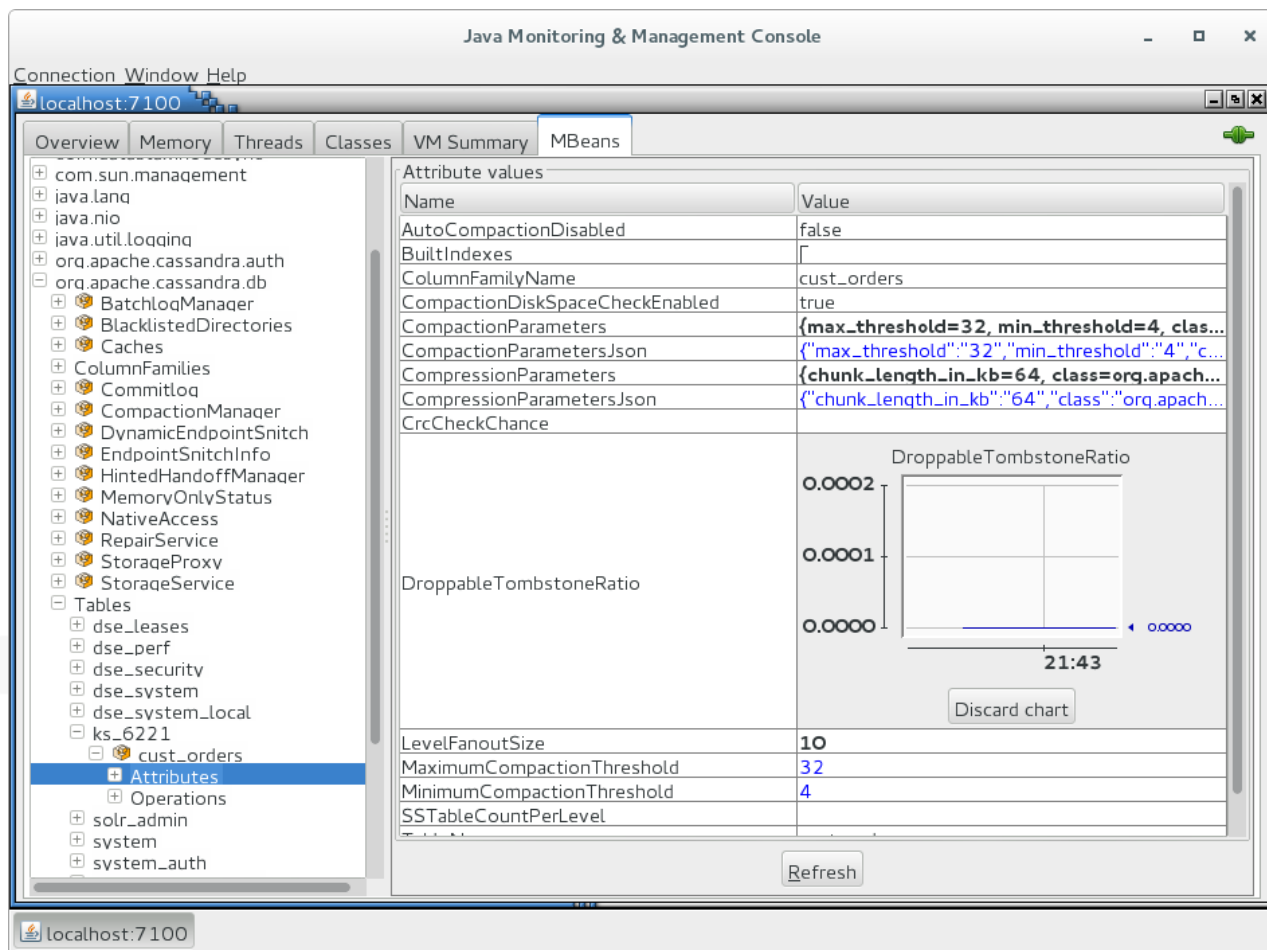
- 6 TABs
- Read/write, graphs
- MBeans TAB most expansive



DSE MBeans

- org.apache.cassandra.db
caching, table metrics, compaction, ..
- org.apache.cassandra.internal
internal server operations; gossip, hinted handoff, ..
- org.apache.cassandra.metrics
client read/write request latencies, Bloom filter false positives, ..
- org.apache.cassandra.net
inter-node comm; StreamingService, MessagingService, FailureDetector , ..
- org.apache.cassandra.request
Tasks related to; read, write, replication, ..
- org.apache.cassandra.service
Data about; node, ring, tokens, snapshots, ..





JConsole/ MBean Examples: Tombstones

Higher ratio is worse

Java Monitoring & Management Console

Connection Window Help

localhost:7100

Overview Memory Threads Classes VM Summary MBeans

Attributes

Name	Value
CompactionHistory	javax.management.openmbean.TabularData...
CompactionSummary	
Compactions	
CoreCompactorThreads	2
CoreValidationThreads	1
CoreViewBuildThreads	2
MaximumCompactorThreads	2
MaximumValidatorThreads	2147483647
MaximumViewBuildThreads	2

Refresh

localhost:7100

JConsole/ MBean Examples: Compaction

Throttling,
MaximumCompactor
Threads
CoreCompactorThre
ads

JConsole/ MBean Examples: Hinted Handoff

3 Hours by default, in
milliseconds

MaxHintsInProgress

Java Monitoring & Management Console

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Overview Memory Threads Classes VM Summary MBeans

com.datastax.bdp.advrep.v2.metric
com.datastax.net
com.datastax.nodesync
com.sun.management
java.lang
java.nio
java.util.logging
org.apache.cassandra.auth
org.apache.cassandra.db
BatchlogManager
BlacklistedDirectories
Caches
ColumnFamilies
CommitLog
CompactionManager
Attributes
Operations
DynamicEndpointSnitch
EndpointSnitchInfo
HintedHandoffManager
MemoryOnlyStatus
NativeAccess
RepairService
StorageProxy
Attributes
Operations
StorageService
Tables
dse_leases
dse_perf
dse_security
dse_system
dse_system_local
ks_6221

Attribute values

Name	Value
CasContentionTimeout	1000
CounterWriteRpcTimeout	5000
CrossDCRttLatency	0
HintedHandoffDisabledDCs	1
HintedHandoffEnabled	true
HintsInProgress	0
IdealConsistencyLevel	Unavailable
MaxHintWindow	10800000
MaxHintsInProgress	256
NativeTransportMaxConcurrentConnections	-1
NumberOfTables	46
OtcBacklogExpirationInterval	0
RangeRpcTimeout	10000
ReadRepairAttempted	0
ReadRepairRepairedBackground	0
ReadRepairRepairedBlocking	0
ReadRpcTimeout	5000
RpcTimeout	10000
SchemaVersions	{ee39bead-4e3d-3b81-accb-2b3a526e7e...
TotalHints	0
TruncateRpcTimeout	60000
WriteRpcTimeout	2000

Refresh

Java Monitoring & Management Console

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Overview Memory Threads Classes VM Summary **MBeans**

com.datastax.bdp.advrep.v2.metric
com.datastax.net
com.datastax.nodesync
com.sun.management
java.lang
java.nio
java.util.logging
org.apache.cassandra.auth
org.apache.cassandra.db
BatchlogManager
BlacklistedDirectories
Caches
ColumnFamilies
Commitlog
CompactionManager
Attributes
Operations
DynamicEndpointSnitch
EndpointSnitchInfo
HintedHandoffManager
MemoryOnlyStatus
NativeAccess
RepairService
StorageProxy
Attributes
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StorageService
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dse_perf
dse_security
dse_system
dse_system_local
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Refresh

localhost:7100

JConsole/ MBean Examples: (Time outs)

RangeRpcTimeout
(range queries)

ReadRpcTimeout
(single read op)

RpcTimeout
(misc inter-node ops)

WriteRpcTimeout
(write ops)

TruncateRpcTimeout
(truncates)

But really: Use nodetool

- nodetool info
Heap and off heap memory, key cache hit rate
- nodetool compactionstats
Status, volume, in-flight
- nodetool tablestats
Index summary usage, memtable usage, read/write latency, partition size, tombstones per read
- nodetool tablehistograms
Table access read/write latencies
- nodetool proxyhistograms
Network read/write latencies



Source: <https://www.pecheur.com/en/ie/buy-node-tool-stonfo-64402.html>

